

PROJECTION-TRANSPORT STEERING: CONDITIONAL CONTROL OF REASONING BEHAVIOR IN LLMs

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Paper under double-blind review

ABSTRACT

Activation steering edits a language model’s residual stream at inference time, but the edit is the same on every input, so a dose that removes a behavior also changes the inputs that never showed it. We introduce Projection-Transport Steering (PTS), which treats a conditional edit as two separate choices: an action that transports a low-dimensional projection of the activation to a target law, and a decision that says when the action runs. We show that the selectivity of any such edit factors exactly into the decision’s error rates and the action’s conversion rates, and that the decision’s share is bounded by its held-out AUROC. The factorization predicts measured selectivity without running the gated edit and identifies detection as the lever. On answer-level overconfidence in multiple-choice reasoning, PTS is more selective than every baseline tuned under the same budget in a pre-registered comparison on two model families, at unchanged accuracy, and it can decide from the prompt or a short prefix in a single pass.

1 INTRODUCTION

Language models carry many behaviors along directions of their activations, and activation steering changes a behavior by moving the activation along such a direction (Turner et al., 2023; Panickssery et al., 2024; Zou et al., 2023; Li et al., 2023; Ardit et al., 2024). The edit needs no training and costs one vector operation per token. It is also applied to every input. A refusal direction added at a useful strength raises refusal on harmless requests as well as harmful ones (Lee et al., 2024), and a behavior that a model shows only on some inputs has to be paid for on all of them.

We study one such behavior, overconfidence in multiple-choice reasoning. The answers to change are the ones the model gets wrong while stating that it is right; the answers to protect are the ones it gets right and says so. The additive edit that removes every overconfident-wrong answer of gemma-2-2b-it also removes every confident-right answer, and weaker doses trade one loss for the other (Sec. 5.2). Conditional methods place a decision in front of the edit: CAST (Lee et al., 2024) conditions on the prompt, and PCHI (Li et al., 2026) conditions a learned head rescaling on a probe. What these methods leave open is how much the decision contributes, what limits it, and when it can be taken.

We introduce Projection-Transport Steering (PTS), which separates the two choices and accounts for each. The action is the cost-minimal edit that moves a projection of the activation to a target law while leaving the orthogonal complement unchanged; it is an optimal-transport map, and the familiar shift, ablation and clamp are its instances for different targets. The decision is a likelihood-ratio test on a statistic of the input. Our method is based on an identity: the selectivity of any conditional edit equals the decision’s true-positive rate times the rate at which the action converts the target state, minus its false-positive rate times the rate at which it converts the protected state. Once the action’s conversion rates are measured, the identity predicts the selectivity of every decision placed in front of it, and it shows that the decision is bounded by how well it separates the two states on questions it has not seen.

Our main contributions are:

method	transfer function $S(p)$	decision (reads)	cost / token
ActAdd, CAA (Turner et al., 2023; Panickssery et al., 2024)	$p + \alpha$	—	$\Theta(d)$
ITI (Li et al., 2023)	$p + \alpha$ on probe-selected heads	—	$\Theta(d)$
directional ablation (Arditi et al., 2024)	0	—	$\Theta(d)$
SAE clamping (Templeton et al., 2024)	c	—	$\Theta(d d_{\text{SAE}})$
MiMiC (Singh et al., 2024)	full-space affine map	—	$\Theta(d^2)$
Linear-AcT (Rodriguez et al., 2025a)	per-neuron affine map	—	$\Theta(d)$
CAST (Lee et al., 2024)	$p + \alpha$	cosine to a condition vector (prompt)	$\Theta(d)$
PCHI (Li et al., 2026)	learned head rescaling	probe (confidence template)	$\Theta(d)$
PTS (ours)	transport map to a target law	likelihood-ratio probe (prompt, prefix or trace)	$\Theta(d)$

Table 1: **Steering operators as a transfer function plus a decision.** Each operator applies a transfer function S to a projection p of the residual stream; the conditional ones add a decision. For $k = 1$, Theorem 1 identifies the cost-minimal S for a target law, and Proposition 1 gives the selectivity of any row as a function of its decision and its action.

- A selectivity identity for conditional edits and a detection ceiling for the decision, together with the characterisation of projection-local actions as transport maps (Sec. 3).
- Evidence that the identity predicts the measured selectivity of gated edits across detectors, actions and thresholds, which lets every gated configuration be selected without running it (Sec. 5.1).
- A pre-registered comparison in which every method is tuned under one rule and evaluated once on 3,000 unseen questions for `gemma-2-2b-it` and `Qwen2.5-7B-Instruct`. PTS is more selective than prompting, additive steering, directional ablation, MiMiC, Linear-AcT and two CAST variants, with accuracy non-inferior at a two-point margin (Sec. 5.2).
- A single-pass operator: the decision reaches nearly its full-trace detection quality from the prompt or from the first tokens of the answer, so PTS runs in one generation at the cost of two dot products per token (Sec. 5.3).

2 RELATED WORK

Global steering. ActAdd (Turner et al., 2023) and CAA (Panickssery et al., 2024) add a contrastive direction to the residual stream, ITI (Li et al., 2023) shifts probe-selected attention heads, RepE (Zou et al., 2023) reads and writes representation directions, directional ablation (Arditi et al., 2024) projects a direction out, and SAE clamping (Templeton et al., 2024) pins a feature. The same edit is applied to every input, and AxBench (Wu et al., 2025) and studies of steering reliability (Tan et al., 2024; Braun et al., 2025) report that such edits are brittle across inputs. Table 1 places these operators in one family of transfer functions.

Distribution matching. MiMiC (Singh et al., 2024) derives the Gaussian optimal-transport affine map between two activation distributions in the full space, Linear-AcT (Rodriguez et al., 2025a) transports each neuron with a univariate map and a strength parameter, and LinEAS (Rodriguez et al., 2025b) learns affine maps end to end. COAST (Nguyen et al., 2026) characterises the covariance-weighted projection that minimises collateral damage for a point target, and A-LQR (Skifstad et al., 2026) treats steering as linear-quadratic control. These results concern the action. Theorem 1 is the projection-local version of the same transport argument, and we do not claim it as the source of the empirical gains; what is new is its pairing with an exact account of what a decision adds.

Conditional steering. CAST (Lee et al., 2024) applies a behavior vector only when the prompt’s hidden state is similar to a condition vector, and is the baseline closest to our design; we reimplement its prompt-level condition and give it the same labelled data as our detector. Dynamic and semantics-adaptive methods scale the edit per token (Ferrando et al., 2026; Wang et al., 2025; Scalena et al., 2024), and GAPS (Haider et al., 2026) gates individual dimensions. Closest in aim is PCHI (Li et al., 2026), which also separates detection from intervention for verbalised overconfidence: a linear probe on the confidence template conditions a learned rescaling of attention heads, reaching large wrong-confident corrections at small damage on mathematics questions. PCHI learns the action for a fixed readout position; we keep the action fixed and ask what the decision contributes, when it can be taken, and how much it can buy. The accounting identity and ceiling of Sec. 3 apply to PCHI as well. Probes read after a few generated tokens predict whether steering will succeed (Fan et al., 2026), detectors of a finished behavior forecast it worse from a prefix (Kortukov et al., 2026), and reading a direction off already-edited activations degrades it (Luo et al., 2026); we measure the prefix cost directly and place the reader above the actor.

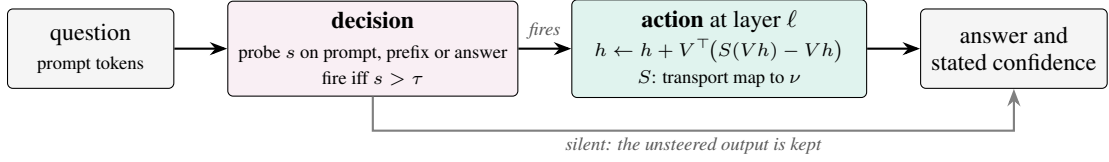


Figure 1: **PTS separates when to edit from how to edit.** A probe reads the input and decides whether to act; the action moves the projection Vh of the residual stream to a target law and leaves the orthogonal complement unchanged. The selectivity of the combination is the decision’s error rates weighted by the action’s conversion rates of the two states (Prop. 1).

3 PROJECTION-TRANSPORT STEERING

A conditional edit makes two choices: how to change an activation, and on which inputs to change it. We call the first the *action* and the second the *decision*. This section fixes the action by a displacement criterion (Sec. 3.1), writes selectivity as a function of the two choices (Sec. 3.2), and describes how the decision is taken in a single pass (Sec. 3.3). Proofs are in Appendix C.

Setup. Let $h \in \mathbb{R}^d$ be the residual-stream activation at a fixed layer and $V \in \mathbb{R}^{k \times d}$ a matrix with orthonormal rows that spans a behavioral subspace. We write $q = Vh$ for the projection, μ for the law of h under unsteered generation, μ_V for the law of q , and ν for a target law on $\mathbb{R}^k$. In our experiments $k = 1$ and $V = \hat{v}^\top$ is the confidence direction, the difference of mean activations between confident and nonconfident answers on an extraction split (Sec. 4).

3.1 THE ACTION

Large edits are what damage generation (Appendix H), so we ask for the smallest edit that moves the projection to a given target law and leaves the rest of the activation alone.

Definition 1 (projection-local steering). *An edit $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is local to V if $(I - V^\top V)T(h) = (I - V^\top V)h$, and it reaches ν if Vh' is distributed as ν when $h \sim \mu$ and $h' = T(h)$. Its cost is $C(T) = \mathbb{E}_\mu \|T(h) - h\|^2$.*

Theorem 1 (local steering is transport). *Let μ_V be absolutely continuous with finite second moment, and let ν have finite second moment. Among local edits that reach ν , the cost is minimised by*

$$T(h) = h + V^\top (S(Vh) - Vh), \quad (1)$$

where S is the optimal-transport (Brenier) map from μ_V to ν ; the minimum cost is $W_2^2(\mu_V, \nu)$, and every minimiser agrees with Eq. (1) on the projection.

Corollary 1 (what a transfer function can and cannot do). *(a) The shift $S(p) = p + \alpha$ is the optimal map iff ν is a translate of μ_V . (b) If two groups of inputs have the same law of p and the event that an input changes state depends on h only through p , every transfer function changes both groups at the same rate.*

In words, once the subspace and the target are fixed, the only freedom left is the one-dimensional map S , and the cheapest S is the monotone rearrangement of quantiles. Table 1 lists the operators this covers. Directional ablation is the map onto δ_0 , a clamp is the map onto a truncated law, and the additive shift $S(p) = p + \alpha$ is the map onto a shifted copy of μ_V . A shift is optimal exactly when the target is a pure translation (Cor. 1); it is the member we end up selecting, and the theorem says it spends no displacement off the confidence coordinate.

3.2 THE DECISION

A decision reads a statistic s of the input and runs the action on the event $\Gamma = \{s > \tau\}$; elsewhere the output is exactly the unsteered one. We measure the result with selectivity, $\text{Sel} = P(\text{leave OCW} \mid \text{OCW}) - P(\text{leave CR} \mid \text{CR})$, the share of overconfident-wrong answers (OCW) the edit removes minus the share of confident-right answers (CR) it destroys. Write TPR and FPR for the rates at which the decision fires on OCW and CR, and ρ_O, ρ_C for the rates at which the action changes the state of a flagged OCW and a flagged CR answer.

Proposition 1 (selectivity accounting). *For any decision and action that leave unflagged outputs unchanged,*

$$\text{Sel} = \text{TPR} \cdot \rho_O - \text{FPR} \cdot \rho_C. \quad (2)$$

Eq. (2) is an identity, and it separates the two choices. The decision sets (TPR, FPR) and the action sets (ρ_O, ρ_C). If the action converts flagged and unflagged answers of the same state at the same rate, ρ_O and ρ_C can be read off an ungated run, and the equation predicts the selectivity of every gated variant before it is run. We use this prediction to select configurations cheaply and test it in Sec. 5.

The decision term is bounded by how well s separates the two states.

Theorem 2 (optimal decision and its ceiling). *(a) For any objective increasing in TPR and decreasing in FPR, including Eq. (2), thresholding the likelihood ratio of s between OCW and CR is optimal at every operating point. (b) If the concave ROC of s has area AUC, then $\max_{\tau} (\text{TPR} - \text{FPR}) \leq 2 \text{AUC} - 1$; for an action with $\rho_O = \rho_C = \rho$ this gives $\text{Sel} \leq \rho (2 \text{AUC} - 1)$.*

We therefore implement the decision as a logistic-regression probe, an estimate of the log-likelihood ratio, fitted on answers labelled by the evaluation readout, and we judge it by its AUROC on questions it was not fitted to. The same bound explains why the decision should not read the coordinate the action edits. On held-out questions the confidence projection separates OCW from CR at AUROC 0.64 for Gemma and 0.53 for Qwen, so a threshold on it buys little, and an edit that depends only on that coordinate changes both states at rates that differ only through the rest of the activation (Cor. 1).

3.3 DECIDING IN ONE PASS

A decision that reads the finished answer requires generating every flagged question twice. The decision can instead read the prompt, or the mean activation over the first t generated tokens, with a probe fitted on exactly that statistic. Theorem 2 applies to any such statistic, so the cost of deciding early is the AUROC lost relative to the finished trace. The reader sits at or above the layer the action writes, the action starts at the decision token, and positions written before it keep their cached keys. The operator then costs two d -dimensional dot products per token.

4 EXPERIMENTAL SETUP

Models and task. The main model is gemma-2-2b-it (Gemma Team, 2024); the confirmatory comparison is repeated on Qwen2.5-7B-Instruct, a second model family, with its own directions, detector and tuning. Both reason inside `<think>...</think>` and answer `\boxed{letter}` on MMLU (Hendrycks et al., 2021); decoding is greedy. The action is applied at layer 14 of 26 (Gemma) and 14 of 28 (Qwen).

Splits. All splits are disjoint sets of MMLU test questions (ids in the released code): an *extraction* split of 400 questions for the directions and transport targets, a *detector* pool of 4,000 for the detector, a *tuning* split of 1,200 on which every method’s configuration is chosen, and a *confirmatory* split of 3,000 used once, under a protocol frozen before its steered rollouts were generated (Appendix A).

What is measured. The primary readout (M5) scores the answer the model actually generates. Correctness is the `\boxed{ }` letter against the gold answer; confidence is $P(\text{YES})$ when the model is then asked, in the same conversation, whether that answer is correct. A steered question is steered for both the reasoning trace and the confidence query, so the answer and its confidence come from the same edited model. Each question is in one of four states: *overconfident-wrong* (OCW, wrong with $P(\text{YES}) \geq 0.5$), *nonconfident-wrong*, *nonconfident-right*, and *confident-right* (CR). Two secondary readouts, the letter-logit confidence of the same answer and a per-option verification readout, are reported in Appendix G; the prompts are in Appendix L.

Metrics. Pairing each question with itself before and after the edit, removal is $P(\text{not OCW after} \mid \text{OCW before})$ over the n_{OCW} questions that started in OCW, retention is $P(\text{CR after} \mid \text{CR before})$ over the n_{CR} that started in CR, and selectivity is $\text{Sel} = \text{removal} - (1 - \text{retention})$: 1 for an edit that removes every OCW and keeps every CR, 0 for one that treats them alike. Δ_{acc} is the change in accuracy over all questions, and ECE is the 10-bin expected calibration error of the confidence.

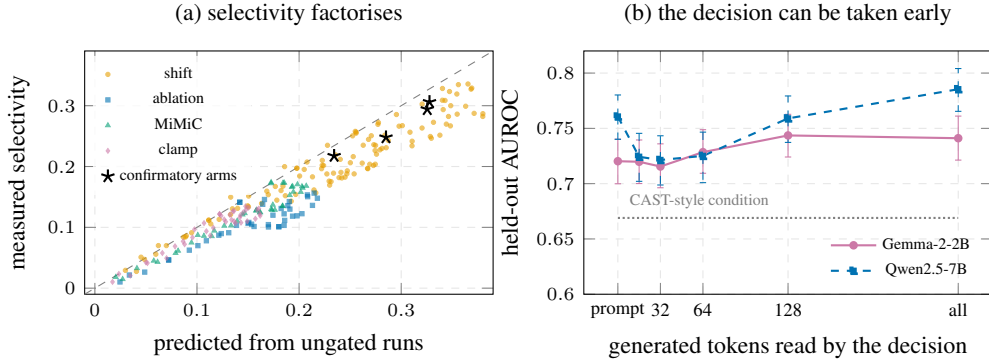


Figure 2: **The decision is the lever, and it can be taken early.** (a) Each point is a decision–action–threshold combination on the Gemma tuning split (240 in total). The horizontal axis is $\text{TPR}_{\rho_O} - \text{FPR}_{\rho_C}$ computed from the ungated run of the action; the vertical axis is the selectivity of the gated edit. Stars are the four gated arms of the confirmatory split, measured on real runs. (b) AUROC between OCW and CR on the 3,000 confirmatory questions of a probe that reads the prompt, the first t generated tokens, or the whole answer; bars are 95% bootstrap intervals, and the dotted line is the trace direction used by the CAST-style condition.

Methods and tuning. We compare prompting (a calibration instruction), additive CAA (Panickssery et al., 2024), directional ablation (Arditi et al., 2024), MiMiC (Singh et al., 2024), Linear-Act (Rodriguez et al., 2025a) and two conditional shifts: CAST with its original prompt condition, a cosine threshold to a condition vector computed from the same labelled pool our detector uses (Lee et al., 2024), and a CAST-style variant conditioned on the finished trace’s projection onto the extraction-split OCW–CR direction. All are reimplemented on the same hooks, records and readouts. The PTS decision is a logistic-regression probe on the mean activation of one layer, fitted on the detector pool, with its layer and regularisation chosen by AUROC on the tuning split; the action is ablation or a shift of the confidence coordinate. Each method searches a grid of 8 to 16 configurations on the tuning split, and one rule selects all of them: the highest selectivity among configurations whose Δacc is at least -0.01 (Appendix E). Post-hoc PTS reads the finished unsteered answer and regenerates the flagged questions; single-pass PTS decides at the prompt, or after $t \in \{16, 32, 64, 128\}$ generated tokens with a probe fitted on that prefix, and acts from then on.

Statistics. The unit of analysis is the question. Intervals are question-level paired bootstrap intervals with 10^4 resamples; comparisons between methods resample the same questions for both. The six primary comparisons of post-hoc PTS against the baselines are Holm-adjusted. We call an accuracy change non-inferior when the lower end of its 95% interval is above -0.02 , a margin fixed in the protocol.

5 RESULTS

5.1 THE SELECTIVITY IDENTITY PREDICTS GATED EDITS

The identity turns one ungated run into a prediction for every decision. On the Gemma tuning split we measure the conversion rates of six actions once, compose each with two decisions at 20 thresholds, and compare the prediction $\text{TPR}_{\rho_O} - \text{FPR}_{\rho_C}$ with the measured selectivity (Fig. 2a). The two agree with correlation 0.97 over 240 combinations; the prediction is higher by 0.033 on average, because the answers a detector flags are slightly harder to convert than the average answer of their state. With the flagged conversion rates in place of the ungated ones the identity is exact, and on the confirmatory split the four gated arms land within 0.010 of the value composed from their ungated runs. We use this to score every gated configuration on the tuning split without running it (Appendix A).

The identity also says what each choice contributes. The shift the selection rule picks for Gemma, $S(p) = p - 0.375\|h\|$, changes the state of 84% of overconfident-wrong and 59% of confident-right answers when applied everywhere. By Prop. 1, a decision recovers 0.59 for every confident-right answer it leaves alone and forfeits 0.84 for every overconfident-wrong answer it misses, so the value of detection grows with how indiscriminate the action is. The probe reads the unsteered answer at held-out AUROC 0.74, against 0.67 for the direction that conditions the CAST-style baseline

method	configs	Δacc [95% CI]	removal $\uparrow$	retention $\uparrow$	Sel $\uparrow$ [95% CI]	ΔSel vs PTS (Holm p)
gemma-2-2b-it, $n = 3000$, $n_{OCW} = 963$, $n_{CR} = 1399$, unsteered accuracy 0.559						
prompting	1	+0.005 [-0.011, +0.020]	0.415	0.785	+0.200 [+0.162, +0.238]	-0.094 [-0.142, -0.045] (<0.001)
additive CAA (Panickssery et al., 2024)	8	-0.001 [-0.016, +0.015]	0.559	0.661	+0.220 [+0.180, +0.259]	-0.074 [-0.124, -0.026] (0.008)
directional ablation (Arditi et al., 2024)	1	+0.009 [-0.007, +0.024]	0.456	0.718	+0.173 [+0.134, +0.212]	-0.121 [-0.172, -0.070] (<0.001)
MiMiC (Singh et al., 2024)	9	+0.000 [-0.015, +0.015]	0.313	0.805	+0.117 [+0.082, +0.154]	-0.177 [-0.226, -0.127] (<0.001)
Linear-AcT (Rodriguez et al., 2025a)	9	+0.005 [-0.005, +0.015]	0.137	0.924	+0.061 [+0.036, +0.088]	-0.233 [-0.278, -0.189] (<0.001)
CAST, prompt condition (Lee et al., 2024)	16	-0.002 [-0.016, +0.012]	0.459	0.758	+0.217 [+0.179, +0.256]	-0.077 [-0.123, -0.031] (0.004)
CAST-style, trace condition	16	-0.005 [-0.017, +0.009]	0.543	0.705	+0.248 [+0.208, +0.288]	-0.046 [-0.085, -0.008] (0.020)
PTS, post-hoc (two passes)	16	+0.003 [-0.009, +0.015]	0.492	0.802	+0.294 [+0.256, +0.334]	—
PTS, decides at token 32 (one pass)	16	-0.008 [-0.019, +0.003]	0.562	0.733	+0.294 [+0.256, +0.333]	+0.000 [-0.033, +0.034]
PTS, decides at the prompt (one pass)	16	-0.012 [-0.025, +0.001]	0.548	0.758	+0.306 [+0.267, +0.345]	+0.012 [-0.024, +0.047]
Qwen2.5-7B-Instruct: pending						

Table 2: **PTS is the most selective method on Gemma at unchanged accuracy.** Confirmatory split of 3,000 MMLU questions, each method at the configuration it won on the tuning split under one rule; removal is over the questions that started overconfident-wrong, retention over those that started confident-right. Intervals are question-level paired bootstrap intervals (10^4 resamples); ΔSel is paired against PTS post-hoc, with p Holm-adjusted over the seven baselines. The single-pass rows are secondary and are compared without adjustment.

(Fig. 2b); at its selected threshold it fires on 54% of overconfident-wrong and 22% of confident-right answers.

5.2 PRE-REGISTERED COMPARISON ON TWO MODEL FAMILIES

A global shift trades one state for the other. On the Gemma tuning split, the shift $p - 0.75\|h\|$ removes every overconfident-wrong answer and every confident-right one, at a cost of 15.3 accuracy points; smaller doses keep more of both (Appendix E). The tuned dose, $p - 0.25\|h\|$, reaches selectivity 0.197 there.

PTS is the most selective method on Gemma. Table 2 reports the frozen configurations on the 3,000 confirmatory questions. PTS reaches selectivity 0.294 [0.256, 0.334], removing 49% of overconfident-wrong answers while keeping 80% of confident-right ones. The strongest baseline, the CAST-style shift conditioned on the trace direction, reaches 0.248, and the paired difference to PTS is 0.046 [0.008, 0.085]; the other six baselines trail by 0.074 to 0.233, and all seven differences remain significant after Holm adjustment. Accuracy changes by +0.003 [-0.009, 0.015], inside the two-point margin, as it does for every baseline. PTS and the tuned additive dose use the same kind of action; the gate is what separates them, with retention 0.80 against 0.66.

On Qwen the identity predicts where the gate helps. @QWENPARA@

The gate also protects unrelated tasks. Applied to every input, the shift $p - 0.5\|h\|$ lowers HumanEval pass@1 from 0.396 to 0.280; with the post-hoc decision in front of it the change is -0.006 [-0.030, 0.012], and with the prompt decision the output is unchanged, because the probe fires on 23% and 0% of coding prompts (Appendix H).

5.3 DECIDING IN ONE PASS

Most of the detection is available before the answer is written. Fig. 2b plots the held-out AUROC of a probe fitted on each statistic. For Gemma the prompt alone reaches 0.720 and the first 16 tokens 0.720, against 0.741 for the finished answer; for Qwen the prompt reaches 0.761 against 0.785. The single-pass operators in Table 2 match the post-hoc one: deciding at token 32 gives selectivity 0.294 and deciding at the prompt 0.306, with paired differences to the post-hoc decision of 0.000 and +0.012. A decision at the prompt costs one dot product on an activation the forward pass already computes, so the operator adds no generation; in an end-to-end measurement of the earlier prefix gate the single-pass operator ran at $1.05\times$ the unsteered wall clock, against $1.97\times$ for regenerating flagged answers (Appendix K).

6 CONCLUSION

Our study has several limitations. First, the evidence covers one behavior on one benchmark family in two instruction-tuned models; an earlier form of the operator passed on one of three models on TruthfulQA, and in the one of six model-behavior cells that reached testing it helped one metric and hurt another (Appendix B). Second, the decision is a probe trained on labelled MMLU answers, and its threshold is tied to that distribution: it fires on part of the open-ended instructions it should leave alone (Appendix H) and too rarely on ARC-Challenge (Appendix J). Third, the transport

characterisation fixes the action for a target law but does not choose the target; we choose it on the tuning split.

We have presented Projection-Transport Steering, which splits a conditional activation edit into an action and a decision and accounts for each. The selectivity identity turns the design of a conditional edit into two measurements, the conversion rates of an action and the held-out error rates of a decision, and in our experiments it is the decision, not the action, that separates the methods.

AI USE STATEMENT

We used generative AI tools to polish the manuscript and to retrieve related work. The authors verified every resulting claim and reference and take responsibility for the final content.

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A CONFIRMATORY PROTOCOL

The protocol was committed to the repository before any steered rollout on the confirmatory split was generated (rebuttal/protocols/confirmatory-v19.md, with two amendments made after the tuning pass and before launch), together with the SHA-256 hashes of the selected configurations and of every fitted detector (confirmatory-v19-gemma.sha256).

Splits. All splits are subsets of the 14,042-question MMLU test set, identified by index and constructed by `label_pool.split_records`. The extraction split (400) excludes the first evaluation subset of the earlier version; the tuning split (1,200) excludes the extraction split and the five earlier evaluation subsets; the detector pool (4,000) and the confirmatory split (3,000) are consecutive blocks of a seeded permutation of the remaining questions. No question appears in two of the four splits.

Selection rule. For every method the configuration with the highest tuning-split selectivity among those with $\Delta_{\text{acc}} \geq -0.01$ is carried unchanged to the confirmatory split. Gated configurations are scored on the tuning split by combining the ungated rollouts of the same split with the gate’s flags (Prop. 1 makes this exact up to batched-decoding nondeterminism; Appendix F); single-pass configurations are real runs. Every confirmatory number is a real run.

Tests. Primary: PTS post-hoc against each of the seven baselines on pooled M5 selectivity, question-level paired bootstrap with 10^4 resamples, two-sided, Holm-adjusted over the seven comparisons. Accuracy: an arm is non-inferior if the lower end of the 95% interval of its accuracy change is above -0.02 . The single-pass and prompt-decision variants, M4 and M2 selectivities, removal, retention and ECE are secondary and reported without a claim.

B AN EARLIER OPERATOR ON OTHER MODELS AND BEHAVIORS

Before the present protocol we froze a simplified operator, an action along a detection direction, after a development analysis on Qwen3, and tested it on held-out TruthfulQA panels of 237 questions (50 in-domain, 187 shifted) for Mistral-7B, OLMo-2-7B and Granite-8B. Each comparison was against a norm-matched spherical action, with 10,000 paired BCa resamples and Bonferroni-adjusted two-sided intervals, and a model passed only if calibrated MC2 and MC1 both improved with intervals excluding zero.

model / panel	comparison	Δ calibrated MC2 [adj. 95% CI]	Δ MC1 [adj. 95% CI]	decision
OLMo-2 / TruthfulQA	matched spherical	+0.137 [+0.105, +0.168]	+0.105 [+0.025, +0.177]	pass
OLMo-2 / TruthfulQA	no intervention	+0.299 [+0.251, +0.345]	+0.287 [+0.177, +0.384]	pass
OLMo-2 / TruthfulQA	strongest isotropic	+0.220 [+0.180, +0.258]	+0.194 [+0.097, +0.287]	pass
OLMo-2 / TruthfulQA	unprompted fit	+0.295 [+0.247, +0.343]	+0.287 [+0.177, +0.384]	pass

Table 3: Held-out OLMo-2 validation on 237 TruthfulQA questions, 50 in-domain and 187 shifted. Intervals use 10,000 paired BCa resamples and the predeclared Bonferroni family; the table shows the matched action and representative construction controls of the seven-control family, all of which pass.

On OLMo-2 the operator passed against the matched action and against all seven pre-registered construction controls (Table 3). Mistral improved calibrated MC2 by +0.153 [+0.108, +0.197], but its MC1 interval $[-0.059, +0.122]$ included zero, and Granite did not clear the screening threshold. A separate screen of the same three models on BBQ and ETHICS, six model-behavior cells with

disjoint fitting, screening and test units, advanced one cell, Mistral on ETHICS, to a 1,000-scenario test. There the operator improved pooled calibrated MC2 by +0.107 [+0.090, +0.124] and lowered MC1 on label-1 scenarios by 0.093 [0.071, 0.124].

C PROOFS AND FORMAL STATEMENTS

Theorem 1. By locality, $T(h) - h \in \text{span}(V^\top) \mu$ -a.s., hence $\|T(h) - h\|^2 = \|VT(h) - Vh\|^2$ because the rows of V are orthonormal, and $C(T) = \mathbb{E}\|VT(h) - Vh\|^2$. The joint law γ of $(Vh, VT(h))$ is a coupling of (μ_V, ν) by the goal constraint, so $C(T) = \int \|y - x\|^2 d\gamma \geq W_2^2(\mu_V, \nu)$. Because μ_V is absolutely continuous and both laws have finite second moments, Brenier’s theorem gives a unique optimal coupling, induced by $S = \nabla\varphi$ with φ convex. $T^*(h) = h + V^\top(S(Vh) - Vh)$ is admissible ($VV^\top = I_k$ gives $VT^*(h) = S(Vh)$, and the complement is untouched) and attains the bound. For any minimiser the induced coupling is optimal, hence equal to the unique one, so $VT(h) = S(Vh)$ a.s. $\square$

For $k = 1$ absolute continuity can be weakened to atomlessness of μ_V : the monotone rearrangement $F_\nu^{-1} \circ F_{\mu_V}$ is then an optimal map for every ν with a finite second moment, including targets with atoms such as δ_0 (ablation) or a law truncated at c (clamp); uniqueness of the map can fail only on a μ_V -null set.

Proposition 2 (metric covariance). *If the cost is $(\Delta h)^\top M(\Delta h)$ with $M \succ 0$, the admissible problem is optimal transport of μ_V to ν in $\mathbb{R}^k$ under the cost $\|\cdot\|_{M_V}^2$, $M_V = VMV^\top \succ 0$, and the optimal operator is $T^*(h) = h + V^\top(S_M(Vh) - Vh)$ with $S_M = M_V^{-1/2} \circ \nabla\varphi \circ M_V^{1/2}$, φ convex.*

Proof. Locality writes $\Delta h = V^\top a$ with $a \in \mathbb{R}^k$, and the goal constraint with $VV^\top = I_k$ fixes $a = S(Vh) - Vh$ for the map S in use. Then $\|\Delta h\|_M^2 = a^\top M_V a$. The cost is a latent-space Mahalanobis cost on $\mathbb{R}^k$; it is not the full-space Mahalanobis distance of Δh unless M commutes with $V^\top V$. In the coordinates $z = M_V^{1/2} q$ it is Euclidean, the change of variables is internal to $\mathbb{R}^k$, and Brenier’s theorem in z (the pushforward of an absolutely continuous law under an invertible linear map is absolutely continuous) gives $\nabla\varphi$; mapping back gives S_M . $\square$

Corollary 1. (a) If $\nu = \mu_V * \delta_\alpha$ the map $x \mapsto x + \alpha$ is the gradient of a convex function pushing μ_V to ν , hence the Brenier map. Conversely, if the optimal map is $x \mapsto x + \alpha$ then ν is its pushforward, the shift. For one-dimensional Gaussians the monotone map is $x \mapsto m_t + (\sigma_t/\sigma_s)(x - m_s)$, with slope σ_t/σ_s . (b) Let E be the event that a trace changes state under the edit and suppose $P(E | h) = g(p, S(p))$. For subpopulation A , $P(E | A) = \mathbb{E}[g(p, S(p)) | A] = \int g(p, S(p)) d\mu_p^A$, and the same for B . If $\mu_p^A = \mu_p^B$ the two integrals coincide. $\square$

The premise of (b) is a conditional-independence assumption: given p , the state change must not depend on the complement $h_\perp$ or on the subpopulation. It is the premise, not the conclusion, that the data reject (Sec. 3).

Proposition 1. Let R be the event that an OCW trace leaves OCW and L the event that a CR trace leaves CR. Off Γ the output equals the unsteered output, which is in its original state by definition, so $R \subseteq \Gamma$ and $L \subseteq \Gamma$ up to null sets. Then $P(R | OCW) = P(\Gamma | OCW)P(R | OCW, \Gamma) = \text{TPR} \rho_O$, and likewise $P(L | CR) = \text{FPR} \rho_C$; subtracting gives Eq. (2). If, within OCW, the indicator that the action converts the trace is independent of Γ , then $\rho_O = P(R^{\text{ungated}} | OCW)$, and likewise for CR; the ungated selectivity is $\rho_O - \rho_C$, and the difference is $\rho_C(1 - \text{FPR}) - \rho_O(1 - \text{TPR})$. $\square$

The identity requires that the unflagged output be exactly the unsteered one. The post-hoc gate satisfies this by construction. The single-pass gate satisfies it for traces on which it never fires, since the action hook is the identity until the decision token.

Theorem 2. (a) is the Neyman–Pearson lemma applied to the class-conditional laws of s : among tests of size at most FPR the likelihood-ratio test maximises TPR, so every objective increasing in TPR and decreasing in FPR is maximised on the likelihood-ratio ROC curve. When s has a monotone likelihood ratio this is a threshold on s . Our detector is a logistic-regression probe, a discriminative estimate of the log-likelihood ratio up to the prior term; (a) states what the probe approximates, not that the fitted probe is optimal. (b) The concavified ROC contains the path

(0, 0) $\rightarrow$ (FPR*, TPR*) $\rightarrow$ (1, 1) through the Youden-optimal point, whose area is $\frac{1}{2}(1 + J)$, so $J \leq 2 \text{AUC} - 1$. With $\rho_O = \rho_C = \rho$, Eq. (2) gives $\text{Sel} = \rho J$. $\square$

Proposition 3 (transport under a fixed decision). *Among interventions local to V , equal to the identity off Γ , and whose projections on Γ follow ν_Γ , the cost-minimal one applies the Brenier map of $(\mu_V | \Gamma) \rightarrow \nu_\Gamma$ on Γ , at cost $P(\Gamma) W_2^2(\mu_V | \Gamma, \nu_\Gamma)$, provided $\mu_V(\cdot | \Gamma)$ is absolutely continuous.*

Proof. Split $C(T)$ over Γ and Γ^c . The second term is zero by the identity constraint; the first, with the conditional goal constraint, is Theorem 1 for $\mu(\cdot | \Gamma)$. $\square$

This result fixes the decision. It does not say that a threshold gate is the best decision for the displacement-priced objective; the next statement shows that it generally is not, and when it is.

Proposition 4 (graded dose). *Let $T_\lambda(h) = h + \lambda(s) V^\top (S(Vh) - Vh)$ with a dose $\lambda : \mathbb{R} \rightarrow [0, 1]$ measurable in the detector statistic s . Write $\pi(s) = P(\text{OCW} | s)$, $d(s) = \mathbb{E}[\|S(Vh) - Vh\|^2 | s] > 0$, and suppose the behavioral gain is linear in the dose, with slope $\beta_1 > 0$ on OCW and $-\beta_0 < 0$ on CR. The dose maximising gain minus displacement priced at $1/(2c)$ is*

$$\lambda^*(s) = \left[\frac{c(\beta_0 + \beta_1)(\pi(s) - \tau)}{d(s)} \right]_0^1, \quad \tau = \frac{\beta_0}{\beta_0 + \beta_1}.$$

It is zero exactly on $\{\pi \leq \tau\}$ and nondecreasing in the ratio $(\pi(s) - \tau)/d(s)$. It is nondecreasing in π when d is constant, but not in general. The threshold gate $\mathbf{1}[\pi > \tau]$ is optimal iff $\lambda^ \in \{0, 1\}$ almost surely, and strictly suboptimal iff $P(0 < \lambda^* < 1) > 0$.*

Proof. The displacement of T_λ is $\lambda(s)^2 \|S(Vh) - Vh\|^2$, with expectation $\mathbb{E}_s[\lambda(s)^2 d(s)]$; the gain is $\mathbb{E}_s[\lambda(s)((\beta_0 + \beta_1)\pi(s) - \beta_0)]$. The objective is an integral over s of a strictly concave quadratic in λ , maximised pointwise on $[0, 1]$ by the clipped stationary point. It is the unique maximiser a.s., so the indicator is optimal iff it agrees with λ^* a.s., i.e. iff $\lambda^* \in \{0, 1\}$ a.s.; otherwise the objective is strictly larger at λ^* on a set of positive measure. For non-monotonicity in π , take s_1, s_2 with $\pi(s_1) = 0.6 < \pi(s_2) = 0.7$, $\tau = 0.5$, $c(\beta_0 + \beta_1) = 1$, $d(s_1) = 0.2$ and $d(s_2) = 1$: then $\lambda^*(s_1) = 0.5 > \lambda^*(s_2) = 0.2$. For a counterexample to strict suboptimality at every finite price, let π take only the values 0.1 and 0.9 with $\tau = 0.5$ and $d \leq 0.4 c(\beta_0 + \beta_1)$: then $\lambda^* \in \{0, 1\}$ a.s. and the threshold gate is optimal. $\square$

Single-pass decision. Let $\mathcal{F}_t$ be generated by the first t generated tokens and z_t the mean detector feature over them. A decision measurable with respect to $\sigma(z_t)$ is a test between the class-conditional laws of z_t , so Theorem 2 applies with z_t in place of s and nothing is assumed about how z_t relates to the finished trace. The reader hook sits at a layer at or above the actor; within one forward pass the actor at position t sees the decision accumulated over positions $< t$, the running mean needs one sum and one counter per sequence, and an edit at position t changes only the keys and values written at t . An earlier version of this paper also stated a time-uniform guarantee for a running-mean gate; it assumed the token contributions form a sub-Gaussian martingale-difference sequence around the finished-trace statistic, which we cannot verify for this statistic, and we no longer use it.

Proposition 5 (computational cost). *Any algorithm implementing a non-degenerate intervention local to a subspace spanned by a dense direction reads all d coordinates of h , so it costs $\Omega(d)$ per token. Eq. (1) costs $\Theta(kd + k \log K)$ with a K -knot lookup, and the detector adds one d -dimensional dot product per token.*

Proof. If a coordinate i with $(\hat{v})_i \neq 0$ is never read, an adversary perturbs h_i , which changes $p = \langle h, \hat{v} \rangle$ and, on a set of positive measure, the correct output, while the algorithm’s output is unchanged. The upper bound counts Vh , a binary search in the lookup, and $h + V^\top(\cdot)$. $\square$

D GSM8K MCQ CONSTRUCTION

Each GSM8K test item with integer gold g receives three deterministic distractors drawn (seeded by item index) from $\{g \pm 1, g \pm 2, g \pm 10, 2g, \lfloor g/2 \rfloor, \text{round}(1.5g), \text{round}(0.8g), g \pm \text{U}\{3..9\}\}$, duplicated, shuffled with the gold into four options. This converts free-form arithmetic into selection while keeping the reasoning burden; we report transfer numbers on it and do not claim GSM8K free-form results.

E TUNING GRIDS

Every configuration searched on the tuning split, with the selected one marked $\star$. Gated rows are scored by composing the ungated rollouts of the same split with the decision’s flags (Appendix F); single-pass rows are real runs. The rule is the highest selectivity among configurations with $\Delta\text{acc} \geq -0.01$, and, when no configuration of a family meets the floor, the highest selectivity overall; this fallback applies to the additive, CAST and prompt-decision families on Qwen, where every shift in the grid costs more than one accuracy point.

Table 4: Tuning split, gemma-2-2b-it ($n = 1,200$, M5).

family	configuration	Δacc	removal	retention	Sel
additive CAA	alpha-0.375	-0.022	0.831	0.424	+0.255
additive CAA	alpha-0.25 $\star$	-0.002	0.536	0.661	+0.197
additive CAA	alpha-0.5	-0.042	0.974	0.136	+0.110
additive CAA	alpha-0.125	+0.010	0.315	0.763	+0.078
additive CAA	alpha-0.75	-0.153	1.000	0.000	+0.000
additive CAA	alpha-1.0	-0.205	1.000	0.000	+0.000
additive CAA	alpha-1.25	-0.270	1.000	0.000	+0.000
additive CAA	alpha-1.5	-0.299	1.000	0.000	+0.000
MiMiC	L10.reg0.0001 $\star$	+0.022	0.349	0.823	+0.172
MiMiC	L10.reg0.1	+0.012	0.349	0.809	+0.158
MiMiC	L10.reg0.01	+0.007	0.320	0.814	+0.135
MiMiC	L18.reg0.1	+0.022	0.260	0.860	+0.121
MiMiC	L18.reg0.01	+0.026	0.242	0.869	+0.111
MiMiC	L18.reg0.0001	+0.023	0.250	0.857	+0.107
MiMiC	L14.reg0.01	+0.001	0.224	0.857	+0.081
MiMiC	L14.reg0.1	+0.004	0.216	0.850	+0.066
MiMiC	L14.reg0.0001	-0.013	0.214	0.841	+0.055
Linear-AcT	L10.lam0.5 $\star$	+0.016	0.133	0.936	+0.069
Linear-AcT	L10.lam1.0	+0.015	0.138	0.919	+0.057
Linear-AcT	L18.lam1.0	+0.012	0.130	0.926	+0.056
Linear-AcT	L14.lam0.25	+0.007	0.128	0.924	+0.052
Linear-AcT	L14.lam1.0	+0.009	0.138	0.913	+0.051
Linear-AcT	L14.lam0.5	+0.010	0.125	0.919	+0.044
Linear-AcT	L18.lam0.5	+0.004	0.117	0.926	+0.043
Linear-AcT	L10.lam0.25	+0.010	0.112	0.919	+0.031
Linear-AcT	L18.lam0.25	+0.005	0.096	0.922	+0.019
CAST-style, trace condition	tau.cr.q30.alpha-0.375	-0.020	0.815	0.458	+0.273
CAST-style, trace condition	tau.cr.q50.alpha-0.375	-0.015	0.732	0.532	+0.264
CAST-style, trace condition	tau.cr.q70.alpha-0.5	-0.014	0.638	0.611	+0.249
CAST-style, trace condition	tau.cr.q70.alpha-0.375 $\star$	-0.006	0.560	0.682	+0.242
CAST-style, trace condition	tau.cr.q70.alpha-0.75	-0.066	0.646	0.595	+0.241
CAST-style, trace condition	tau.cr.q50.alpha-0.5	-0.028	0.859	0.380	+0.239
CAST-style, trace condition	tau.cr.q50.alpha-0.75	-0.110	0.875	0.353	+0.228
CAST-style, trace condition	tau.cr.q90.alpha-0.5	-0.001	0.411	0.802	+0.214
CAST-style, trace condition	tau.cr.q90.alpha-0.75	-0.022	0.414	0.795	+0.209
CAST-style, trace condition	tau.cr.q90.alpha-0.375	+0.000	0.370	0.832	+0.202
CAST-style, trace condition	tau.cr.q30.alpha-0.25	-0.001	0.521	0.670	+0.190
CAST-style, trace condition	tau.cr.q50.alpha-0.25	+0.000	0.474	0.716	+0.190
CAST-style, trace condition	tau.cr.q30.alpha-0.5	-0.038	0.945	0.244	+0.189
CAST-style, trace condition	tau.cr.q70.alpha-0.25	+0.000	0.352	0.809	+0.161
CAST-style, trace condition	tau.cr.q30.alpha-0.75	-0.139	0.969	0.178	+0.147
CAST-style, trace condition	tau.cr.q90.alpha-0.25	+0.001	0.245	0.894	+0.139
CAST, prompt condition	L16.q50.alpha-0.375	-0.018	0.607	0.673	+0.280
CAST, prompt condition	L16.q30.alpha-0.375	-0.017	0.711	0.565	+0.276
CAST, prompt condition	L14.q30.alpha-0.375	-0.017	0.708	0.567	+0.275
CAST, prompt condition	L14.q50.alpha-0.375	-0.018	0.594	0.680	+0.274
CAST, prompt condition	L16.q50.alpha-0.5	-0.022	0.682	0.580	+0.262
CAST, prompt condition	L14.q50.alpha-0.5	-0.021	0.669	0.588	+0.258
CAST, prompt condition	L16.q50.alpha-0.75	-0.069	0.690	0.562	+0.252
CAST, prompt condition	L14.q50.alpha-0.75	-0.070	0.674	0.571	+0.245
CAST, prompt condition	L14.q30.alpha-0.5	-0.034	0.810	0.415	+0.225
CAST, prompt condition	L16.q30.alpha-0.5	-0.034	0.815	0.408	+0.223
CAST, prompt condition	L14.q30.alpha-0.75	-0.106	0.820	0.378	+0.198
CAST, prompt condition	L16.q30.alpha-0.75	-0.104	0.826	0.369	+0.195
CAST, prompt condition	L16.q30.alpha-0.25 $\star$	-0.007	0.443	0.730	+0.172
CAST, prompt condition	L14.q30.alpha-0.25	-0.008	0.440	0.728	+0.168
CAST, prompt condition	L16.q50.alpha-0.25	-0.009	0.362	0.792	+0.153
CAST, prompt condition	L14.q50.alpha-0.25	-0.010	0.354	0.792	+0.146
PTS post-hoc	q60.alpha-0.75	-0.031	0.570	0.763	+0.334
PTS post-hoc	q50.alpha-0.75	-0.050	0.674	0.654	+0.328
PTS post-hoc	q40.alpha-0.75	-0.073	0.771	0.555	+0.326
PTS post-hoc	q30.alpha-0.75	-0.093	0.862	0.454	+0.316
PTS post-hoc	q30.alpha-0.375	-0.013	0.737	0.578	+0.315
PTS post-hoc	q60.alpha-0.375 $\star$	-0.003	0.510	0.790	+0.300
PTS post-hoc	q40.alpha-0.375	-0.015	0.654	0.645	+0.299
PTS post-hoc	q50.alpha-0.375	-0.012	0.586	0.707	+0.293

family	configuration	Δacc	removal	retention	Sel
PTS post-hoc	q30.alpha-0.25	+0.002	0.477	0.735	+0.212
PTS post-hoc	q40.alpha-0.25	+0.002	0.424	0.767	+0.191
PTS post-hoc	q50.alpha-0.25	+0.002	0.380	0.806	+0.186
PTS post-hoc	q60.alpha-0.25	+0.005	0.320	0.864	+0.184
PTS post-hoc	q30.ablate	+0.012	0.388	0.765	+0.153
PTS post-hoc	q40.ablate	+0.014	0.336	0.795	+0.131
PTS post-hoc	q50.ablate	+0.016	0.281	0.841	+0.122
PTS post-hoc	q60.ablate	+0.012	0.232	0.887	+0.119
PTS, prefix decision	q50.at32.alpha-0.375*	+0.004	0.573	0.737	+0.310
PTS, prefix decision	q50.at16.alpha-0.375	-0.002	0.573	0.731	+0.304
PTS, prefix decision	q30.at16.alpha-0.375	+0.002	0.729	0.572	+0.302
PTS, prefix decision	q50.at64.alpha-0.375	+0.017	0.562	0.739	+0.301
PTS, prefix decision	q30.at32.alpha-0.375	+0.002	0.690	0.602	+0.293
PTS, prefix decision	q50.at128.alpha-0.375	-0.001	0.547	0.708	+0.255
PTS, prefix decision	q30.at64.alpha-0.375	+0.010	0.656	0.595	+0.252
PTS, prefix decision	q30.at128.alpha-0.375	+0.000	0.651	0.574	+0.225
PTS, prefix decision	q50.at32	+0.019	0.245	0.894	+0.139
PTS, prefix decision	q30.at32	+0.018	0.320	0.816	+0.137
PTS, prefix decision	q50.at64	+0.002	0.214	0.898	+0.111
PTS, prefix decision	q30.at16	-0.002	0.326	0.781	+0.106
PTS, prefix decision	q50.at16	+0.003	0.224	0.878	+0.102
PTS, prefix decision	q50.at128	+0.000	0.182	0.912	+0.094
PTS, prefix decision	q30.at64	-0.001	0.250	0.836	+0.086
PTS, prefix decision	q30.at128	-0.003	0.214	0.871	+0.085
PTS, prompt decision	q60.alpha-0.75	-0.041	0.570	0.767	+0.337
PTS, prompt decision	q60.alpha-0.5*	-0.009	0.568	0.769	+0.336
PTS, prompt decision	q50.alpha-0.75	-0.061	0.656	0.661	+0.317
PTS, prompt decision	q50.alpha-0.5	-0.018	0.654	0.663	+0.316
PTS, prompt decision	q60.alpha-0.375	-0.015	0.518	0.790	+0.308
PTS, prompt decision	q50.alpha-0.375	-0.010	0.589	0.712	+0.301
PTS, prompt decision	q30.alpha-0.375	-0.013	0.724	0.569	+0.293
PTS, prompt decision	q40.alpha-0.5	-0.024	0.724	0.567	+0.291
PTS, prompt decision	q40.alpha-0.75	-0.084	0.729	0.562	+0.291
PTS, prompt decision	q40.alpha-0.375	-0.012	0.646	0.643	+0.289
PTS, prompt decision	q30.alpha-0.5	-0.020	0.826	0.463	+0.288
PTS, prompt decision	q30.alpha-0.75	-0.099	0.839	0.445	+0.284
PTS, prompt decision	q30.alpha-0.25	+0.007	0.464	0.735	+0.199
PTS, prompt decision	q60.alpha-0.25	-0.002	0.320	0.871	+0.191
PTS, prompt decision	q40.alpha-0.25	+0.004	0.409	0.781	+0.190
PTS, prompt decision	q50.alpha-0.25	+0.002	0.365	0.825	+0.190
earlier gate + ablation	u.crq50.ablate*	+0.007	0.378	0.740	+0.118
ungated actions	ablate	+0.007	0.451	0.691	+0.141
ungated actions	clamp.q50	+0.017	0.307	0.814	+0.122
ungated actions	clamp.q30	+0.012	0.339	0.763	+0.102
ungated actions	otq.cal	+0.004	0.115	0.926	+0.040

Table 5: Tuning split, Qwen2.5-7B-Instruct ($n = 1,200$, M5).

family	configuration	Δacc	removal	retention	Sel
additive CAA	alpha-0.125*	-0.026	0.212	0.911	+0.123
additive CAA	alpha-0.375	-0.050	0.208	0.892	+0.100
additive CAA	alpha-0.25	-0.044	0.200	0.887	+0.087
additive CAA	alpha-0.5	-0.076	0.204	0.855	+0.059
additive CAA	alpha-0.75	-0.138	0.233	0.759	-0.007
additive CAA	alpha-1.0	-0.217	0.412	0.544	-0.043
additive CAA	alpha-1.25	-0.334	0.567	0.272	-0.161
additive CAA	alpha-1.5	-0.473	0.375	0.230	-0.395
MiMiC	L18.reg0.0001	-0.013	0.279	0.906	+0.185
MiMiC	L18.reg0.01*	-0.007	0.263	0.911	+0.173
MiMiC	L14.reg0.01	-0.018	0.254	0.912	+0.166
MiMiC	L14.reg0.0001	-0.023	0.233	0.912	+0.145
MiMiC	L10.reg0.1	-0.019	0.250	0.890	+0.140
MiMiC	L14.reg0.1	-0.021	0.217	0.923	+0.140
MiMiC	L18.reg0.1	-0.013	0.233	0.906	+0.139
MiMiC	L10.reg0.01	-0.022	0.237	0.892	+0.129
MiMiC	L10.reg0.0001	-0.031	0.233	0.868	+0.101
Linear-AcT	L18.lam1.0*	-0.004	0.192	0.948	+0.140
Linear-AcT	L14.lam1.0	+0.002	0.175	0.955	+0.130
Linear-AcT	L14.lam0.25	+0.004	0.138	0.969	+0.106
Linear-AcT	L10.lam1.0	-0.008	0.158	0.947	+0.105
Linear-AcT	L18.lam0.25	-0.006	0.138	0.957	+0.095
Linear-AcT	L10.lam0.25	-0.013	0.133	0.958	+0.092
Linear-AcT	L18.lam0.5	-0.006	0.142	0.950	+0.091
Linear-AcT	L10.lam0.5	-0.013	0.125	0.951	+0.076
Linear-AcT	L14.lam0.5	-0.005	0.113	0.961	+0.073
CAST-style, trace condition	tau.cr.q30.alpha-0.375	-0.043	0.188	0.907	+0.094
CAST-style, trace condition	tau.cr.q30.alpha-0.25	-0.035	0.183	0.906	+0.089
CAST-style, trace condition	tau.cr.q50.alpha-0.375	-0.040	0.158	0.918	+0.076

	family	configuration	Δ acc	removal	retention	Sel
702						
703	CAST-style, trace condition	tau.cr.q50.alpha-0.25	-0.033	0.154	0.919	+0.074
704	CAST-style, trace condition	tau.cr.q30.alpha-0.5	-0.067	0.188	0.879	+0.067
	CAST-style, trace condition	tau.cr.q70.alpha-0.375	-0.024	0.108	0.945	+0.053
705	CAST-style, trace condition	tau.cr.q50.alpha-0.5	-0.059	0.154	0.898	+0.052
706	CAST-style, trace condition	tau.cr.q70.alpha-0.25	-0.022	0.104	0.945	+0.049
	CAST-style, trace condition	tau.cr.q50.alpha-0.75	-0.079	0.188	0.858	+0.045
707	CAST-style, trace condition	tau.cr.q90.alpha-0.25	-0.010	0.067	0.975	+0.041
708	CAST-style, trace condition	tau.cr.q70.alpha-0.5	-0.036	0.104	0.937	+0.041
	CAST-style, trace condition	tau.cr.q90.alpha-0.375*	-0.009	0.062	0.979	+0.041
709	CAST-style, trace condition	tau.cr.q90.alpha-0.5	-0.018	0.071	0.970	+0.041
	CAST-style, trace condition	tau.cr.q70.alpha-0.75	-0.048	0.125	0.912	+0.037
710	CAST-style, trace condition	tau.cr.q30.alpha-0.75	-0.098	0.212	0.824	+0.036
711	CAST-style, trace condition	tau.cr.q90.alpha-0.75	-0.023	0.079	0.956	+0.035
712	CAST, prompt condition	L14.q30.alpha-0.25*	-0.032	0.167	0.916	+0.082
	CAST, prompt condition	L16.q30.alpha-0.25	-0.032	0.167	0.916	+0.082
713	CAST, prompt condition	L14.q30.alpha-0.375	-0.040	0.163	0.918	+0.081
714	CAST, prompt condition	L16.q30.alpha-0.375	-0.040	0.163	0.918	+0.081
	CAST, prompt condition	L16.q50.alpha-0.375	-0.034	0.142	0.931	+0.072
715	CAST, prompt condition	L14.q50.alpha-0.375	-0.034	0.142	0.929	+0.071
716	CAST, prompt condition	L14.q50.alpha-0.25	-0.026	0.138	0.932	+0.069
	CAST, prompt condition	L16.q50.alpha-0.25	-0.027	0.133	0.932	+0.065
717	CAST, prompt condition	L14.q30.alpha-0.5	-0.058	0.167	0.890	+0.057
718	CAST, prompt condition	L16.q30.alpha-0.5	-0.058	0.167	0.890	+0.057
	CAST, prompt condition	L16.q50.alpha-0.5	-0.050	0.133	0.914	+0.048
719	CAST, prompt condition	L14.q50.alpha-0.5	-0.051	0.133	0.913	+0.046
720	CAST, prompt condition	L16.q50.alpha-0.75	-0.078	0.158	0.868	+0.026
	CAST, prompt condition	L14.q50.alpha-0.75	-0.079	0.150	0.868	+0.018
721	CAST, prompt condition	L14.q30.alpha-0.75	-0.105	0.188	0.821	+0.009
722	CAST, prompt condition	L16.q30.alpha-0.75	-0.105	0.183	0.822	+0.006
723	PTS post-hoc	q40.ablate*	-0.006	0.254	0.927	+0.181
	PTS post-hoc	q50.ablate	-0.002	0.233	0.940	+0.173
724	PTS post-hoc	q30.ablate	-0.009	0.254	0.918	+0.172
	PTS post-hoc	q60.ablate	-0.003	0.212	0.948	+0.161
725	PTS post-hoc	q40.alpha-0.375	-0.044	0.183	0.907	+0.090
726	PTS post-hoc	q30.alpha-0.375	-0.048	0.192	0.898	+0.090
	PTS post-hoc	q40.alpha-0.25	-0.043	0.183	0.902	+0.085
727	PTS post-hoc	q60.alpha-0.375	-0.032	0.154	0.929	+0.084
728	PTS post-hoc	q30.alpha-0.25	-0.045	0.192	0.892	+0.083
	PTS post-hoc	q60.alpha-0.25	-0.028	0.150	0.931	+0.081
729	PTS post-hoc	q50.alpha-0.375	-0.039	0.163	0.916	+0.078
730	PTS post-hoc	q50.alpha-0.25	-0.037	0.154	0.912	+0.066
	PTS post-hoc	q60.alpha-0.75	-0.062	0.158	0.883	+0.041
731	PTS post-hoc	q50.alpha-0.75	-0.083	0.179	0.854	+0.033
732	PTS post-hoc	q40.alpha-0.75	-0.098	0.204	0.826	+0.030
	PTS post-hoc	q30.alpha-0.75	-0.110	0.217	0.805	+0.021
733	PTS, prefix decision	q30.at32	-0.010	0.163	0.951	+0.113
734	PTS, prefix decision	q30.at32.alpha-0.375	-0.012	0.183	0.929	+0.113
	PTS, prefix decision	q30.at16*	-0.008	0.163	0.948	+0.111
735	PTS, prefix decision	q50.at32	-0.007	0.142	0.966	+0.108
736	PTS, prefix decision	q50.at16	-0.006	0.138	0.960	+0.097
	PTS, prefix decision	q30.at64	-0.008	0.133	0.951	+0.084
737	PTS, prefix decision	q30.at16.alpha-0.375	-0.030	0.167	0.916	+0.082
738	PTS, prefix decision	q50.at32.alpha-0.375	-0.013	0.133	0.943	+0.077
	PTS, prefix decision	q30.at64.alpha-0.375	-0.014	0.133	0.942	+0.075
739	PTS, prefix decision	q50.at64	-0.005	0.108	0.963	+0.072
740	PTS, prefix decision	q30.at128.alpha-0.375	-0.005	0.083	0.969	+0.052
	PTS, prefix decision	q50.at128.alpha-0.375	-0.004	0.079	0.972	+0.051
741	PTS, prefix decision	q50.at16.alpha-0.375	-0.029	0.113	0.938	+0.051
742	PTS, prefix decision	q50.at64.alpha-0.375	-0.018	0.083	0.955	+0.038
	PTS, prefix decision	q30.at128	+0.002	0.058	0.979	+0.037
743	PTS, prefix decision	q50.at128	+0.002	0.050	0.977	+0.027
744	PTS, prompt decision	q30.alpha-0.375*	-0.045	0.204	0.901	+0.105
	PTS, prompt decision	q30.alpha-0.25	-0.042	0.196	0.899	+0.095
745	PTS, prompt decision	q40.alpha-0.375	-0.043	0.188	0.907	+0.094
746	PTS, prompt decision	q50.alpha-0.375	-0.040	0.171	0.916	+0.086
	PTS, prompt decision	q40.alpha-0.25	-0.043	0.175	0.906	+0.081
747	PTS, prompt decision	q50.alpha-0.25	-0.038	0.163	0.914	+0.077
748	PTS, prompt decision	q60.alpha-0.375	-0.033	0.138	0.935	+0.072
	PTS, prompt decision	q30.alpha-0.5	-0.068	0.204	0.868	+0.072
749	PTS, prompt decision	q60.alpha-0.25	-0.031	0.133	0.929	+0.063
	PTS, prompt decision	q50.alpha-0.5	-0.056	0.163	0.897	+0.059
750	PTS, prompt decision	q40.alpha-0.5	-0.068	0.175	0.874	+0.049
751	PTS, prompt decision	q60.alpha-0.5	-0.047	0.129	0.914	+0.044
752	PTS, prompt decision	q50.alpha-0.75	-0.080	0.179	0.855	+0.034
	PTS, prompt decision	q60.alpha-0.75	-0.067	0.146	0.884	+0.030
753	PTS, prompt decision	q30.alpha-0.75	-0.108	0.217	0.810	+0.026
754	PTS, prompt decision	q40.alpha-0.75	-0.101	0.196	0.826	+0.022
	earlier gate + ablation	u.crq50.ablate*	-0.008	0.196	0.938	+0.134
755	ungated actions	ablate	-0.008	0.258	0.913	+0.171
	ungated actions	clamp.q30	-0.015	0.221	0.918	+0.139

family	configuration	Δ_{acc}	removal	retention	Sel
ungated actions	otq-cal	-0.003	0.154	0.962	+0.116
ungated actions	clamp-q50	-0.008	0.183	0.928	+0.112

F THE SELECTIVITY IDENTITY IN PRACTICE

Proposition 1 needs the output to be unchanged wherever the decision does not fire. For the post-hoc and prompt decisions this holds by construction, so a gated arm equals, question by question, either its unsteered row or its ungated row. Composing the two requires one ungated run per action and one set of decision scores, and it differs from a real gated run only through batched-decoding nondeterminism: a flagged question is generated in a different batch in the two cases. Table 6 compares the composition with the real run for the four gated arms of the confirmatory split, and reports the prediction from the ungated conversion rates (independence within each state).

arm (Gemma, confirmatory)	TPR	FPR	ρ_O	ρ_C	predicted	composed	real run
PTS post-hoc	0.541	0.222	0.841	0.585	+0.325	+0.296	+0.294
PTS at the prompt	0.549	0.242	0.981	0.872	+0.328	+0.306	+0.306
CAST-style, trace	0.603	0.380	0.841	0.585	+0.285	+0.244	+0.248
CAST, prompt	0.805	0.635	0.559	0.339	+0.234	+0.207	+0.217

Table 6: Real gated runs against the composition of ungated runs and against the prediction $\text{TPR } \rho_O - \text{FPR } \rho_C$ with ungated conversion rates.

The prediction is higher than the measurement in all four rows, by 0.017 to 0.037. The same bias appears on the tuning split (Fig. 2a): the answers a detector flags are converted less often than the average answer of their state, so the independence assumption of Prop. 1 holds only approximately. The composition, which uses no such assumption, is what scores the gated configurations during selection.

G SECONDARY READOUTS

Table 7 reports the Gemma confirmatory arms under the two secondary readouts. The letter-logit confidence M2 assigns confidence above 0.5 to nearly every answer, so its OCW stratum is close to the set of wrong answers and every method moves it little. Under M4 the edits act on four separate verification generations; PTS is level with the ungated edits there, which is expected for a decision fitted to M5 labels.

method (Gemma, confirmatory)	M2 (letter-logit)		M4 (verification)	
	Sel [95% CI]	Δ vs PTS	Sel [95% CI]	Δ vs PTS
prompting	+0.083 [+0.052, +0.113]	-0.008 [-0.045, +0.028]	+0.138 [+0.086, +0.191]	-0.102 [-0.167, -0.037]
additive CAA (Panickssery et al., 2024)	+0.089 [+0.059, +0.120]	-0.001 [-0.036, +0.034]	+0.242 [+0.187, +0.296]	+0.003 [-0.061, +0.067]
directional ablation (Arditi et al., 2024)	+0.092 [+0.062, +0.122]	+0.002 [-0.034, +0.036]	+0.246 [+0.193, +0.299]	+0.007 [-0.054, +0.068]
MiMiC (Singh et al., 2024)	+0.064 [+0.036, +0.092]	-0.027 [-0.063, +0.008]	+0.244 [+0.189, +0.298]	+0.004 [-0.059, +0.067]
Linear-AcT (Rodriguez et al., 2025a)	+0.029 [+0.009, +0.050]	-0.061 [-0.091, -0.031]	+0.111 [+0.068, +0.155]	-0.128 [-0.185, -0.071]
CAST, prompt condition (Lee et al., 2024)	+0.073 [+0.045, +0.102]	-0.017 [-0.052, +0.017]	+0.207 [+0.155, +0.259]	-0.032 [-0.092, +0.028]
CAST-style, trace condition	+0.074 [+0.046, +0.103]	-0.016 [-0.041, +0.010]	+0.224 [+0.173, +0.277]	-0.015 [-0.063, +0.033]
PTS, post-hoc (two passes)	+0.090 [+0.064, +0.118]	—	+0.239 [+0.192, +0.286]	—
PTS, decides at the prompt (one pass)	+0.099 [+0.069, +0.128]	+0.008 [-0.023, +0.038]	+0.245 [+0.198, +0.292]	+0.006 [-0.039, +0.051]

Table 7: Secondary readouts on the confirmatory split; paired bootstrap intervals, no multiplicity adjustment.

H CAPABILITY AWAY FROM THE TARGET BEHAVIOR

Table 8 measures code synthesis (HumanEval, 164 problems, greedy, pass@1), open-ended instruction following (200 AlpacaEval prompts, 384 new tokens; distinct-3 and the unsteered model’s perplexity of the continuation) and language modelling (300 WikiText-103 passages, teacher forced). Gated rows use the frozen Gemma decisions at the frozen threshold; each gated output is the unsteered output where the decision is silent and the ungated output where it fires. The coding and open-ended prompts are outside the distribution the probes were fitted on, and the decisions fire on some of them: the post-hoc probe on 23% of HumanEval, 42% of AlpacaEval and all WikiText

method	HumanEval		AlpacaEval prompts		WikiText ppl
	pass@1	Δ [95% CI]	distinct-3	fluency ppl	
unsteered	0.396	+0.000 [+0.000, +0.000]	0.953	3.31	25.5
shift -0.375 (ungated)	0.384	-0.012 [-0.073, +0.049]	0.889	3.72	28.1
PTS post-hoc, shift -0.375	0.402	+0.006 [+0.000, +0.018]	0.909	3.49	28.1
PTS prompt, shift -0.375	0.396	+0.000 [+0.000, +0.000]	0.906	3.49	25.8
shift -0.5 (ungated)	0.281	-0.116 [-0.183, -0.049]	0.815	4.00	32.5
PTS post-hoc, shift -0.5	0.390	-0.006 [-0.030, +0.012]	0.865	3.55	32.5
PTS prompt, shift -0.5	0.396	+0.000 [+0.000, +0.000]	0.866	3.61	26.2

Table 8: Capability under the Gemma operators. Intervals are paired bootstrap intervals over problems.

passages, the prompt probe on 0%, 47% and 12%. Where it fires on open-ended text the shift costs distinct-3 and fluency as it does ungated.

@OODPARA@

I EXAMPLES

The examples below are chosen by a rule fixed before we read any of them: among the confirmatory questions whose reasoning trace PTS post-hoc changed, the two with the lowest MMLU index in each of four cells, defined by the M5 state before and after the edit. The cells hold 474 (overconfident-wrong, state changed), 47 (overconfident-wrong, unchanged), 33 (confident-right, kept) and 277 (confident-right, lost) questions; confident-right questions the decision does not flag are not in these counts and keep their unsteered output. Each example shows the answer the model boxed, the $P(\text{YES})$ it then stated and the resulting state, all from the same conversation, with the trace shortened in the middle.

overconfident-wrong, state changed (mmlu-4, gold B).

Question: Find the product of the given polynomials in the given polynomial ring. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$. A. $2x^2 + 5$ B. $6x^2 + 4x + 6$ C. 0 D. $x^2 + 1$

before (answer C, $P(\text{YES}) = 0.78$, OCW): $\langle \text{think} \rangle$ The product of two polynomials in a polynomial ring is found by multiplying them together. We need to consider the remainders when each term is divided by 8. Let's multiply the polynomials and then reduce each term modulo 8. $\langle \text{think} \rangle f(x) * g(x) = (4x - 5)(2x^2 - 4x + 2) = 8x^3 - 16x^2 + 8x - 10x^2 + 20x - 10 = 8x^3 - 26x^2 + 28x - 10$ $\langle \text{think} \rangle$ Now we need to reduce each term modulo 8. $\langle \text{think} \rangle 8x^3 - 26x^2 + 28x - 10 \equiv 0x^3 - 2x^2 + 0x - 2 \pmod{8}$ $\langle \text{think} \rangle$ The product of the polynomials is -2. $\langle \text{think} \rangle \boxed{C}$

after (answer A, $P(\text{YES}) = 0.01$, NCW): $\langle \text{think} \rangle$ The product of the polynomials is $f(x)g(x) = (4x - 5)(2x^2 - 4x + 2)$ in $\mathbb{Z}_8[x]$. We can expand this using the distributive property. $\langle \text{think} \rangle$ The expansion is $(4x - 5)(2x^2 - 4x + 2) = 8x^3 - 16x^2 + 8x - 10x^2 + 20x - 10 = 8x^3 - 26x^2 + 28x - 10$ $\langle \text{think} \rangle$ Now we need to reduce each term modulo 8. $\langle \text{think} \rangle 8x^3 - 26x^2 + 28x - 10 = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1)$

overconfident-wrong, state changed (mmlu-23, gold D).

Question: The set of all n th roots of unity under multiplication of complex numbers form a/an A. semi group with identity B. commutative semigroups with identity C. group D. abelian group

before (answer C, $P(\text{YES}) = 0.98$, OCW): $\langle \text{think} \rangle$ The n th roots of unity are complex numbers that satisfy the equation $z^n = 1$. We need to check if this set satisfies the properties of a group. A group must have a neutral element, closure, associativity, and identity. $\langle \text{think} \rangle$ The set of n th roots of unity satisfies the properties of a group. It has a neutral element ($z^n = 1$) and closure. $\langle \text{think} \rangle$ The set of n th roots of unity is commutative. $\langle \text{think} \rangle$ The set of n th roots of unity is abelian. $\langle \text{think} \rangle \boxed{C}$

after (answer C, $P(\text{YES}) = 0.30$, NCW): $\langle \text{think} \rangle$ The set of all n th roots of unity is a set of complex numbers. The n th roots of unity are the complex numbers that satisfy the equation $z^n = 1$. Since the set is closed under multiplication, it is a semigroup. The identity element is 1. The set is also closed under multiplication. The set is commutative. The set is abelian. $\langle \text{think} \rangle \boxed{C}$

overconfident-wrong, state unchanged (mmlu-57, gold B).

Question: Find the generator for the finite field $\mathbb{Z}_{11}$. A. 1 B. 2 C. 3 D. 4

before (answer A, $P(\text{YES}) = 1.00$, OCW): $\langle \text{think} \rangle$ The generator of a finite field is an element that generates all other elements in the field. $\mathbb{Z}_{11}$ is a finite field with 11 elements. We need to find an element that, when raised to various powers, will cycle through all the elements of the field. We can start by looking at the powers of 1. $\langle \text{think} \rangle$ The powers of 1 in $\mathbb{Z}_{11}$ are: $1, 1^2 = 1, 1^3 = 1, \dots$ $\langle \text{think} \rangle$ This shows that 1 is a generator for $\mathbb{Z}_{11}$. $\boxed{A}$

after (answer A, $P(\text{YES}) = 0.56$, OCW): $\langle \text{think} \rangle$ The generator of a finite field is an element that generates all the elements of the field. Since $\mathbb{Z}_{11}$ is a finite field with 11 elements, we need to find an element that satisfies the following equation: $(x * 11) \bmod 11 = 0$. $\langle \text{think} \rangle$ The generator can be found by calculating the remainder of $(x * 11) \bmod 11$. $\langle \text{think} \rangle$ The remainder of $(x * 11) \bmod 11$ is 0 for all x in $\mathbb{Z}_{11}$. $\langle \text{think} \rangle$ Therefore, the generator is 1. $\boxed{A}$

overconfident-wrong, state unchanged (mmlu-1888, gold C).

Question: Suppose that we are interested in testing the null hypothesis that a GARCH(2,2) model can be restricted to a process with a constant conditional variance using the likelihood ratio test approach. Which of the following statements are true? A. The test statistic will follow a chi-squared distribution with 2 degrees of freedom under the null hypothesis B. The value of the log-likelihood function will almost always be bigger for the restricted model than for the unrestricted model C. If the relevant values of the log-likelihood functions are -112.3 and -118.4, the value of the test statistic is 12.2 D. The likelihood ratio test compares the slopes of the log-likelihood function at the maximum and at the restricted parameter value.

before (answer A, $P(\text{YES}) = 0.92$, OCW): $\langle \text{think} \rangle$ The null hypothesis is that the GARCH(2,2) model can be restricted to a process with a constant conditional variance. This means we are testing if the model has a non-constant variance. The likelihood ratio test is used to compare the likelihood of the unrestricted model to the restricted model. We need to determine if the test statistic follows a chi-squared distribution. $\langle \text{think} \rangle$ **A. The test statistic will [...] the unrestricted and restricted models. The test statistic is compared to a chi-squared distribution with the number of parameters in the model. $\langle \text{think} \rangle$ **A**

after (answer A, $P(\text{YES}) = 0.53$, OCW): $\langle \text{think} \rangle$ The null hypothesis is that the GARCH(2,2) model can be restricted to a process with a constant conditional variance. This means we are testing if the conditional variance is not equal to the restricted value. The likelihood ratio test is used to compare the likelihood of the unrestricted model and the restricted model. The test statistic will follow a chi-squared distribution with 2 degrees of freedom under the null hypothesis. $\boxed{A}$

confident-right, kept (mmlu-196, gold A).

Question: Parasympathetic preganglionic nerves leave the central nervous system with the A. third cranial nerves. B. fourth cranial nerves. C. fifth cranial nerves. D. sixth cranial nerves.

before (answer A, $P(\text{YES}) = 0.96$, CR): $\langle \text{think} \rangle$ The parasympathetic nervous system is responsible for "rest and digest" functions. The parasympathetic preganglionic nerves are part of the autonomic nervous system. The autonomic nervous system is divided into the sympathetic and parasympathetic branches. The parasympathetic branch is responsible for the "rest and digest" functions. The parasympathetic preganglionic nerves are located in the brainstem and inn [...] rves. The cranial nerves are numbered I-XII. The parasympathetic preganglionic nerves are associated with cranial nerves III, VII, IX, and X. $\boxed{A}$

after (answer A, $P(\text{YES}) = 0.68$, CR): $\langle \text{think} \rangle$ The parasympathetic nervous system is responsible for the "rest and digest" functions. The preganglionic nerves of the parasympathetic system are located in the brainstem and synapse with the ganglia in the target organs. The target organs are the same as the parasympathetic ganglia. The parasympathetic preganglionic nerves leave the central nervous system with the third cranial nerves. $\boxed{A}$

confident-right, kept (mmlu-1430, gold A).

Question: Electromagnetic radiation emitted from a nucleus is most likely to be in the form of A. gamma rays B. microwaves C. ultraviolet radiation D. visible light

before (answer A, $P(\text{YES}) = 1.00$, CR): $\langle \text{think} \rangle$ The nucleus of an atom is a very small, dense region containing protons and neutrons. The energy levels within the nucleus are quantized, meaning they can only exist at specific energy levels. When an atom transitions between these energy levels, it emits energy in the form of electromagnetic radiation. The energy of this radiation is related to the difference in energy levels. Higher energy levels correspond [...] high energy and short wavelength. Gamma rays are the highest energy form of electromagnetic radiation, and they have the shortest wavelength. $\boxed{A}$

after (answer A, $P(\text{YES}) = 0.78$, CR): $\langle \text{think} \rangle$ The nucleus is a very small and dense

object. The energy of the nucleus is very high. The energy of the nucleus is emitted in the form of gamma rays. $\boxed{A}$

confident-right, lost (mmlu-6, gold A).

Question: Statement 1 — Every homomorphic image of a group G is isomorphic to a factor group of G . Statement 2 — The homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G . A. True, True B. False, False C. True, False D. False, True

before (answer A, $P(\text{YES}) = 0.85$, CR): $\langle \text{think} \rangle$ Statement 1 says that every homomorphic image of a group G is isomorphic to a factor group of G . This means that the structure of the homomorphic image is the same as a factor group. Statement 2 says that the homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G . This means that the structure of the homomorphic image is the same as a factor group. $\langle \text{think} \rangle$ $\boxed{A}$

after (answer A, $P(\text{YES}) = 0.03$, NCR): $\langle \text{think} \rangle$ Statement 1 is true. The homomorphic image of a group G is isomorphic to a factor group of G . Statement 2 is also true. The homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G . $\langle \text{think} \rangle$ $\boxed{A}$

confident-right, lost (mmlu-31, gold B).

Question: If $(G, \cdot)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}$, for all a, b in G , then G is a/an A. commutative semi group B. abelian group C. non-abelian group D. None of these

before (answer B, $P(\text{YES}) = 0.96$, CR): $\langle \text{think} \rangle$ The given condition $(ab)^{-1} = a^{-1}b^{-1}$ implies that the group G is closed under the operation. This means that the operation is associative. Also, the inverse of a product is the product of the inverses. This is a characteristic of abelian groups. $\langle \text{think} \rangle$ $\boxed{B}$

after (answer A, $P(\text{YES}) = 0.11$, NCW): $\langle \text{think} \rangle$ The given condition $(ab)^{-1} = a^{-1}b^{-1}$ implies that the group G is commutative. The commutative group is a group where $a*b = b*a$ for all a, b in G . $\langle \text{think} \rangle$ $\boxed{A}$

J TRANSFER TO OTHER BENCHMARKS

We ran the Gemma operators, with the directions, probes and thresholds fitted on MMLU, on 300 ARC-Challenge questions and 300 GSM8K questions converted to four-option multiple choice (Appendix D). These runs are exploratory and not part of the protocol. Table 9 shows that the MMLU threshold does not carry to ARC: the post-hoc probe fires on 59 of 300 questions and PTS reaches selectivity 0.100, against 0.370 for the ungated additive dose. On GSM8K PTS keeps 0.944 of confident-right answers at selectivity 0.287, while the additive dose reaches 0.319 at retention 0.634.

method	ARC-Challenge ($n_{OCW} = 66, n_{CR} = 205$)			GSM8K-MCQ ($n_{OCW} = 35, n_{CR} = 161$)		
	Δ_{acc}	retention	Sel	Δ_{acc}	retention	Sel
additive CAA (tuned)	+0.027	0.810	+0.370	+0.023	0.634	+0.319
CAST-style, trace condition	-0.003	0.863	+0.060	-0.060	0.516	+0.173
CAST, prompt condition	-0.017	0.844	+0.223	+0.023	0.634	+0.319
PTS, post-hoc	-0.010	0.888	+0.100	+0.023	0.944	+0.287
PTS, decides at the prompt	-0.013	0.941	-0.013	-0.040	0.814	+0.128

Table 9: Transfer of the Gemma operators fitted on MMLU, M5 readout, 300 questions each, no refitting.

K RESULTS UNDER THE EARLIER PROTOCOL

An earlier version of this work used the verification readout M4 (Appendix L) as its primary instrument, five evaluation subsets of 300 MMLU questions, a detector read off the extraction split, and configurations tuned on a 1,200-question split. We keep its results here as a record, with three qualifications that the confirmatory protocol removes. Under M4 the edited generations are the per-option verification traces, not the reasoning trace, so these results describe the verification task. Four of the five subsets share 7 to 14 questions with the extraction split (38 in total); the pooled numbers below drop them. The subsets also overlap each other in 54 questions, which are counted once.

Pooled comparison. Table 10 pools the five subsets (1,406 unique questions, 245 OCW and 380 CR under M4) and pairs every method with the tuned gated ablation of that protocol. No tuned baseline is above it at the 5% level, and none is below it either except the single-pass gate; the

method (earlier protocol, tuned)	M4		M2	
	Sel [95% CI]	Δ vs gated abl.	Sel [95% CI]	Δ acc
gated ablation (tuned)	+0.203 [+0.125, +0.282]	—	+0.042 [+0.006, +0.078]	−0.005
additive CAA (tuned)	+0.271 [+0.188, +0.350]	+0.067 [−0.024, +0.156]	+0.082 [+0.039, +0.127]	−0.009
directional ablation	+0.258 [+0.178, +0.338]	+0.054 [−0.017, +0.126]	+0.080 [+0.038, +0.124]	−0.004
CAST-style (tuned)	+0.221 [+0.146, +0.294]	+0.017 [−0.075, +0.108]	+0.101 [+0.060, +0.143]	−0.005
MiMiC (tuned)	+0.126 [+0.050, +0.203]	−0.077 [−0.180, +0.023]	+0.044 [+0.006, +0.082]	−0.006
Linear-AcT (tuned)	+0.146 [+0.082, +0.211]	−0.057 [−0.155, +0.039]	+0.032 [+0.001, +0.064]	−0.004
prompting	+0.196 [+0.118, +0.273]	−0.008 [−0.112, +0.096]	+0.053 [+0.011, +0.096]	−0.011
score-proportional dose (tuned)	+0.241 [+0.163, +0.320]	+0.038 [−0.026, +0.100]	+0.057 [+0.017, +0.099]	−0.008
sequential gate (tuned)	+0.079 [+0.021, +0.138]	−0.124 [−0.215, −0.031]	+0.024 [−0.004, +0.052]	−0.003
gated ablation (published)	+0.278 [+0.201, +0.356]	+0.075 [+0.005, +0.146]	+0.076 [+0.036, +0.117]	+0.004
additive −0.75 (published)	+0.000 [+0.000, +0.000]	−0.203 [−0.282, −0.125]	−0.025 [−0.078, +0.028]	−0.171

Table 10: Earlier protocol, five subsets pooled with overlapping questions removed. Intervals are question-level paired bootstrap intervals (10^4 resamples).

tuned additive dose has the highest mean (+0.271 against +0.203, paired difference +0.067, 95% CI [−0.024, +0.156]). Under M2 on the same traces every selectivity is below 0.11.

Earlier detector. The two-dimensional LDA gate of the earlier version reached AUROC 0.77–0.80 between OCW and CR on the 400 extraction traces it was fitted to, and 0.59 (layer 14) and 0.59 (layer 16) on the 1,406 pooled evaluation questions; the one-dimensional detection direction reached 0.66. This in-sample optimism is what motivated fitting the detector on a separate pool and reporting only held-out AUROC.

Scale, layers, geometry and cost. Table 11 repeats the published configurations on gemma-2-9b-it; Table 12 refits the axes and the gate at nine depths; Table 13 reports displacement, nearest-neighbour distance, the fraction of tokens outside the 99% conditional ellipsoid and next-token KL; Table 14 gives end-to-end wall clock on one H100 for 300 questions.

method (M4, 9B, 3 subsets)	Δ acc	ECE	ocw_rm	cr_keep	AURC	Sel
baseline	acc 0.633 $\pm$ 0.031	0.144 $\pm$ 0.021	—	—	0.197 $\pm$ 0.025	—
additive −0.75	−0.184 $\pm$ 0.033	0.167 $\pm$ 0.037	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.450 $\pm$ 0.023	+0.000 $\pm$ 0.000
CAST-style gate+add	−0.114 $\pm$ 0.032	0.101 $\pm$ 0.011	0.831 $\pm$ 0.058	0.453 $\pm$ 0.038	0.285 $\pm$ 0.016	+0.283 $\pm$ 0.093
clamp q_{50} (ours)	+0.023 $\pm$ 0.012	0.143 $\pm$ 0.018	0.315 $\pm$ 0.012	0.887 $\pm$ 0.015	0.184 $\pm$ 0.025	+0.202 $\pm$ 0.020
gated ablation (ours)	+0.011 $\pm$ 0.008	0.127 $\pm$ 0.009	0.297 $\pm$ 0.026	0.907 $\pm$ 0.014	0.186 $\pm$ 0.023	+0.203 $\pm$ 0.034

Table 11: gemma-2-9b-it, layer 24, published configurations, mean $\pm$ standard deviation over three subsets of 300 questions.

layer	Cohen’s d conf.	Cohen’s d OCW/CR	gate AUROC	2AUC−1
4	+0.525	−0.079	0.783	+0.566
7	+0.535	−0.093	0.756	+0.511
10	+0.512	−0.144	0.741	+0.482
12	+0.516	−0.150	0.742	+0.485
14	+0.589	−0.155	0.769	+0.538
16★	+0.631	−0.141	0.799	+0.599
18	+0.616	−0.116	0.779	+0.558
22	+0.675	−0.110	0.791	+0.582
25	+0.700	−0.111	0.649	+0.298

Table 12: Layer sweep on the 400 extraction traces (in-sample AUROC).

L PROMPTS

All prompts are rendered with the model’s chat template. Text in braces is filled per question; everything else is verbatim.

Reasoning prompt (all readouts).

Given the following multiple choice question, choose the single best answer (A, B, C, or D).
Please think step-by-step and put your thinking inside
<think>...</think> tokens.
After </think>, give ONLY your final answer in the format:

operator	MMLU				ARC				GSM8K			
	disp.	NN	out%	KL	disp.	NN	out%	KL	disp.	NN	out%	KL
unsteered	—	1.00	0.9	0.000	—	1.00	1.2	0.000	—	1.00	2.9	0.000
random direction, norm-matched	0.663	1.90	1.6	0.445	0.644	2.27	1.7	0.233	0.749	2.06	0.9	0.149
additive -0.75 Panickssery et al. (2024)	0.663	1.88	100.0	0.032	0.644	2.25	100.0	0.008	0.749	2.04	100.0	0.012
additive, tuned dose Panickssery et al. (2024)	0.221	1.22	69.9	0.005	0.215	1.33	40.3	0.001	—	—	—	—
MiMiC Singh et al. (2024)	0.082	1.06	1.3	0.001	0.074	1.08	1.8	0.000	0.097	1.06	4.5	0.001
Linear-AcT Rodriguez et al. (2025a)	0.012	1.01	1.0	0.000	0.011	1.01	1.5	0.000	0.012	1.01	3.5	0.000
clamp q_{50} (ours)	0.034	1.03	1.3	0.006	0.046	1.06	1.1	0.001	0.003	1.00	2.9	0.001
quantile transport (ours)	0.004	1.00	1.0	0.000	0.003	1.00	1.3	0.000	0.007	1.00	4.0	0.000
ablation Arditì et al. (2024)	0.144	1.15	29.5	0.010	0.178	1.28	28.4	0.001	0.062	1.05	10.1	0.001

Table 13: Activation diagnostics at layer 14 under teacher forcing of unsteered traces. NN is the nearest-neighbour distance to 4,096 natural activations relative to the unsteered value; out% is the share of tokens outside the 99% conditional ellipsoid of the edited coordinate given the leading 64 principal directions of the complement (1% expected); KL is the mean next-token divergence. These are measured distances, not a test of membership in a manifold.

method	passes	seconds	wall clock
unsteered	1	43.5	1.00×
additive -0.75 Panickssery et al. (2024)	1	45.2	1.04×
ablation Arditì et al. (2024)	1	44.6	1.02×
clamp q_{50} (ours)	1	41.4	0.95×
quantile transport (ours)	1	44.8	1.03×
MiMiC Singh et al. (2024)	1	42.9	0.98×
Linear-AcT Rodriguez et al. (2025a)	1	40.9	0.94×
gated ablation, post-hoc gate (ours) (223/300 regenerated)	2	85.8	1.97×
gated ablation, sequential gate (ours)	1	45.5	1.05×

Table 14: End-to-end wall clock for 300 MMLU questions, batch 64, one H100, one run per method.

```

\boxed{[letter]}
Question: {question}
Options:
A. {option A}
B. {option B}
C. {option C}
D. {option D}

```

M5 (primary). After the model’s reply, a second user turn is added to the same conversation and $P(\text{YES})$ is read from the next-token logits, summing the probabilities of the single-token variants of YES and of NO and renormalising over the two:

```

You answered {letter}. Is that answer correct? Reply with
only YES or NO.

```

The letter is the one with the highest logit after the last `\boxed{` of the trace, which is appended when the trace has none (6.1% of unsteered confirmatory traces). On the unsteered confirmatory split this letter equals the written letter on 96.9% of questions; the remainder are traces with several boxes or none. A question is confident iff $P(\text{YES}) \geq 0.5$.

M2 (secondary). The confidence of the same generated answer is the softmax over the four letter logits at the position after `\boxed{`.

M4 (secondary; primary in the earlier version). Each option is verified in a separate generation, with at most 192 new tokens:

```

Answer whether the given answer is the right answer for the
following question.
Please think step-by-step and put your thinking inside
<think>...</think> tokens.
After </think>, answer ONLY with "YES" or "NO".
Question:
{question}
Answer:
{option text}

```

$P(\text{YES})$ is read at the answer word after `</think>`; the predicted answer is the option with the largest $P(\text{YES})$ and its confidence is that value divided by the sum over options. When an M4 condition is steered, the hook is active in these four verification generations, not in the reasoning trace, so an M4 result describes the verification task.

Prompting baseline. The instruction below is prepended to the user turn (Gemma has no system role) or passed as the system message (Qwen):

```
Be careful and well-calibrated. Only answer confidently
when you are sure; if you are uncertain, explicitly
acknowledge the uncertainty in your reasoning and
confidence.
```

Weights. Gemma-2-2B weights were served from the `unsloth/gemma-2-2b-it` mirror, whose 288 tensors are bit-identical to `google/gemma-2-2b-it`; Qwen weights are Qwen/Qwen2.5-7B-Instruct. Decoding is greedy with at most 320 new tokens for the reasoning trace.